AIDAN NGUYEN

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SUMMARY

Bioengineering student (UC Berkeley, Fall 2026) at the intersection of machine learning and biology. Two years of wet lab research experience in vascular biology and Drosophila genetics combined with independent deep learning projects in protein function prediction. Motivated by applying ML to drug discovery — building toward a research career at the interface of computation and biology.

EDUCATION

B.S. Bioengineering | *University of California, Berkeley* — *Expected May 2028*
Transfer student, Fall 2026

A.S. Transfer — Bioengineering | *Gavilan College, Gilroy CA* — *Graduating May 2026*
GPA: 3.73

TECHNICAL SKILLS

ML & Programming	Python · PyTorch · scikit-learn · NumPy · pandas · ESM-2/protein language models · Git/GitHub · Google Colab
Wet Lab	Cell culture (HUVECs) · Drosophila husbandry & dissection · Gel electrophoresis · PCR · Holotomographic microscopy
Languages	English (primary) · Spanish (fluent)

RESEARCH EXPERIENCE

Research Assistant — Jurney Lab | *San Jose State University* — *Summer 2025*

– Cultured HUVECs under static and flow conditions using an Ibidi pump system to model physiological wall shear stress; developed proficiency in live-cell maintenance and passage protocols.
– Designed and implemented a manual annotation pipeline for segmenting mitochondrial networks from holotomographic microscopy image sequences, generating binary image datasets for downstream computational analysis.

Research Assistant — Grillo-Hill Lab | *San Jose State University* — *Summer 2024*

– Maintained Drosophila melanogaster lines and supported optimization of temperature-shift protocols (GAL4-UAS system) to induce controlled tumorigenic overgrowth in wing imaginal discs.

PROJECTS

Protein Function Classifier (V2) | *Deep Learning — In Progress* — *2026*

– Building a multitask deep learning classifier predicting enzyme commission (EC) class and Gene Ontology (GO) biological process terms from raw amino acid sequence using pretrained ESM-2 transformer embeddings.
– Benchmarking against CLEAN using homology-aware CD-HIT train/test splits to ensure genuine generalization rather than sequence memorization; evaluating macro-F1 across rare and common enzyme classes.

Protein Function Classifier (V1) | *Engineering Programming — Course Project* — *Spring 2025*

– Built a scikit-learn classifier predicting protein function from sequence-derived features; first end-to-end ML pipeline connecting raw sequence data to biological function annotation.

ADDITIONAL EXPERIENCE

STEM Peer Tutor | *Gavilan College* *August 2024 – Present*

– Provide embedded tutoring for calculus and physics across multiple sections; develop study guides and facilitate exam review sessions.
– Apply evidence-based techniques including the Socratic Method to build student critical thinking and academic independence.

LEADERSHIP & ACTIVITIES

Founder | *Gavilan Golden Mean Philosophical Society* *December 2024 – Present*

– Founded and grew a campus philosophy club; organize weekly Socratic discussions on ethics, existentialism, and applied philosophy.